MAE 155A: Aerospace Engineering Design I

Effective technical presentations

Winter 2026

John Hwang
Associate Professor

UC San Diego
JACOBS SCHOOL OF ENGINEERING
Mechanical and Aerospace Engineering

Gradient-based optimization

UC San Diego
JACOBS SCHOOL OF ENGINEERING
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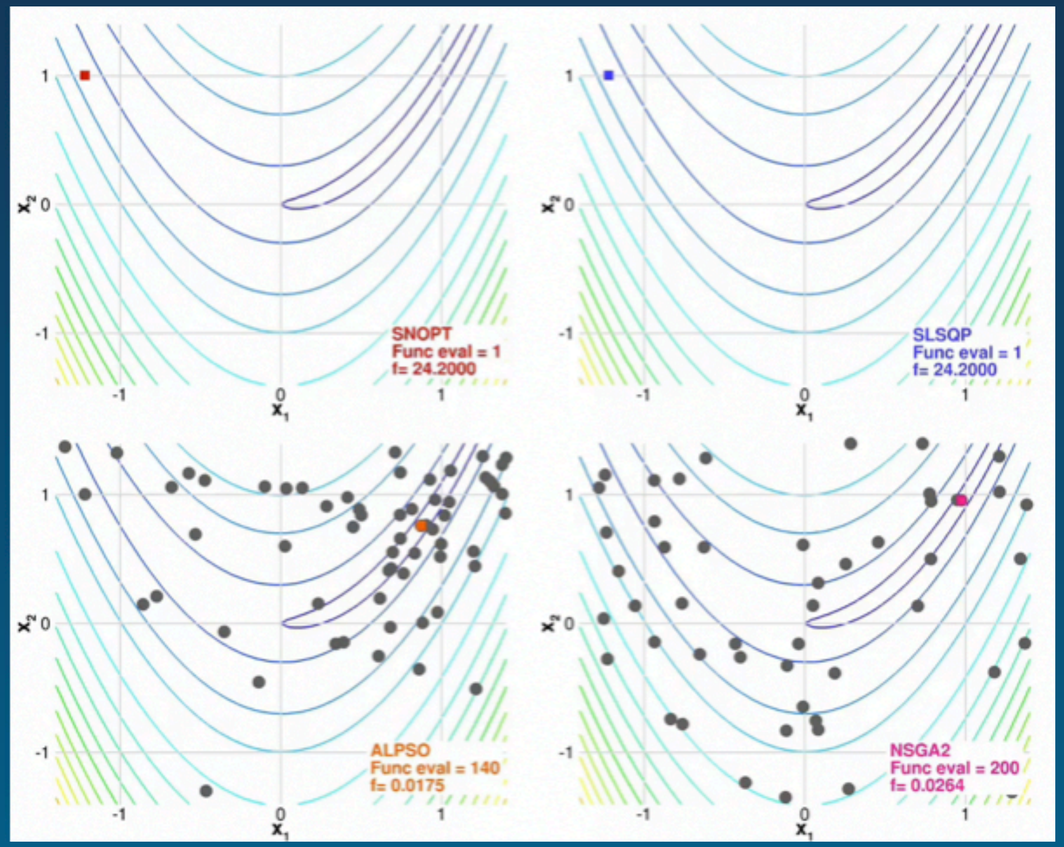


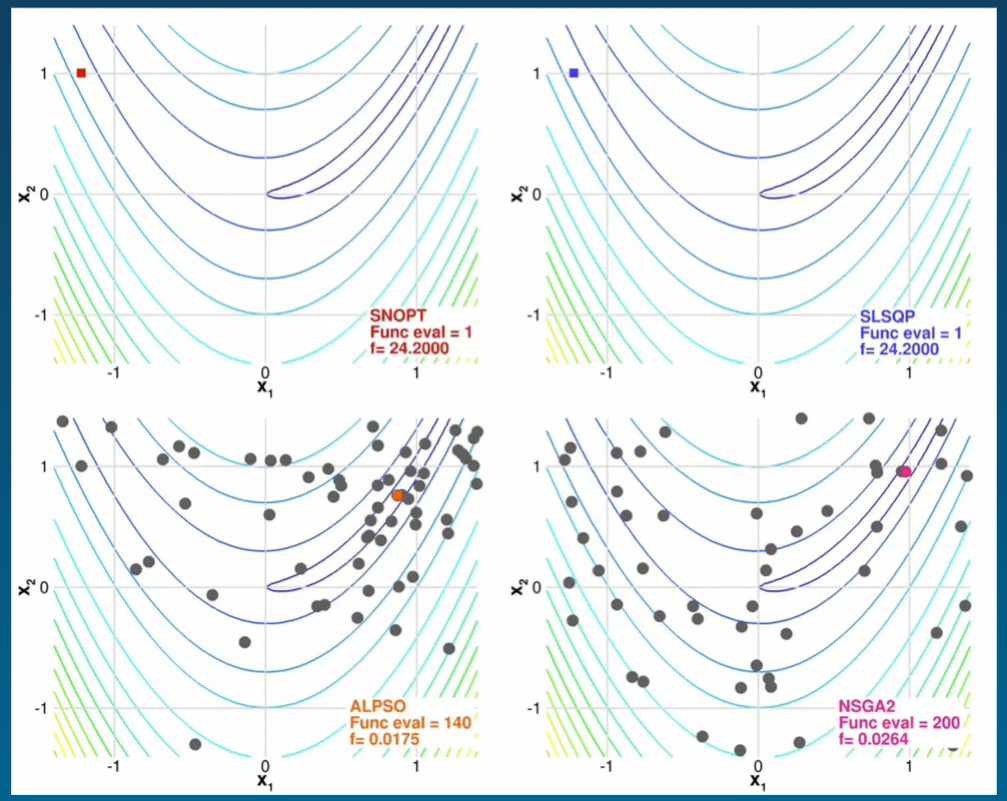
Gradient-based optimization algorithms are able to rapidly converge by taking into account derivative information, consisting of the gradient of the objective and the constraint Jacobian. However, when gradient-based optimization algorithms are used, there are additional challenges that must be overcome and are not present in gradient-free optimization. These challenges are:

- The presence of **multiple local minima** and the **greater sensitivity of the optimum** found by the gradient-based optimization algorithm to the initial design variable vector.
- The problems encountered when the objective function or constraint function is a **discontinuous function.**
- Similar problems encountered when the objective function or constraint function is a **non-differentiable function.**
- Dealing with cases in which the **model fails to converge** — since the algorithm updates the design variable vector in each iteration, a failed model evaluation may cause the algorithm to stall because it cannot continue from the current design point.

However, in practice, these challenges do not diminish the value of gradient-based optimization.

Improved

Gradient-based optimization has several challenges but each challenge is addressable



Additional challenges:

Multiple local minima

The goal is to improve the design

Discontinuous functions

Fixable using smooth transitions

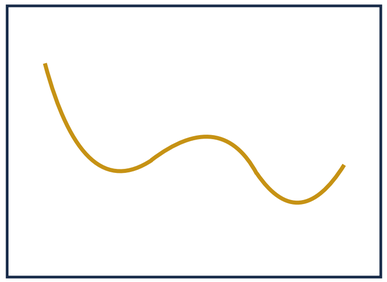
Non-differentiable functions

Fixable using smooth transitions

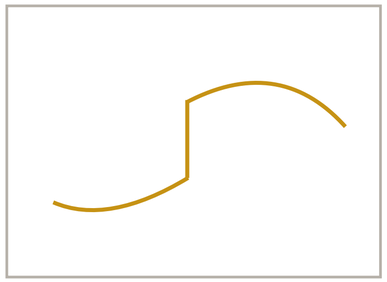
Model convergence failures

Can be handled in a backtracking line search

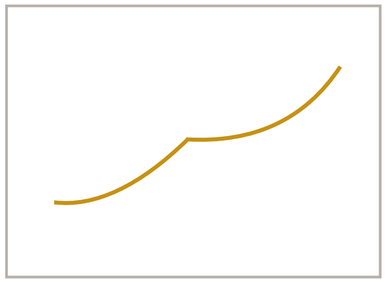
Gradient-based optimization has several challenges but each challenge is addressable



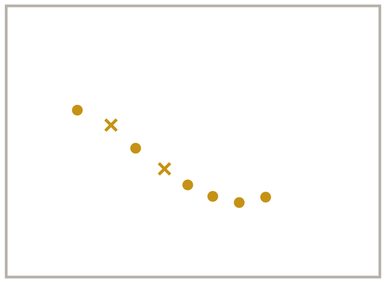
Multiple
local minima



Discontinuous
functions



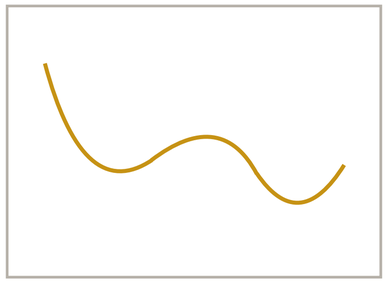
Non-differentiable
functions



Model convergence
failures

The goal is to improve the design; global optimizer is infeasible

Gradient-based optimization has several challenges but each challenge is addressable



Multiple
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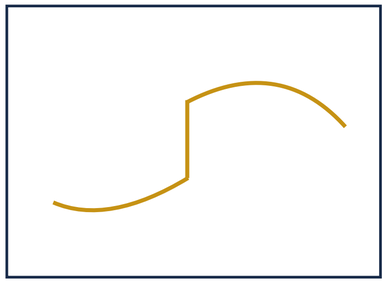
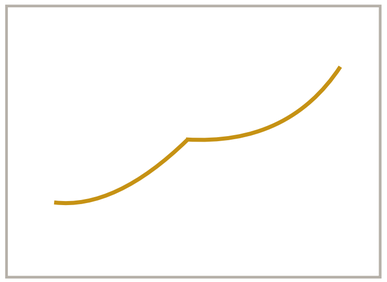
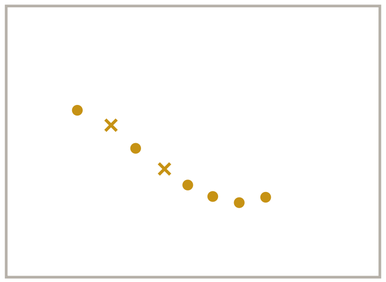


Discontinuous
functions

Fixable using a smooth transition (e.g., a cubic function) or curve fit

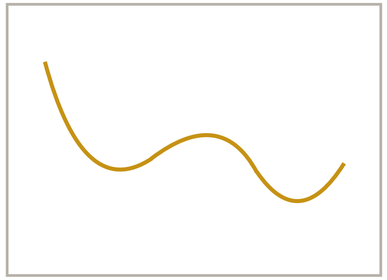


Non-differentiable
functions



Model convergence
failures

Gradient-based optimization has several challenges but each challenge is addressable



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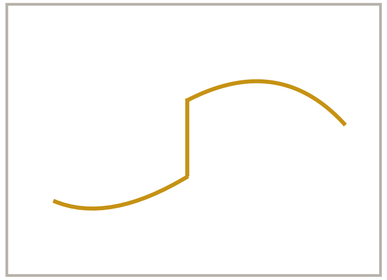

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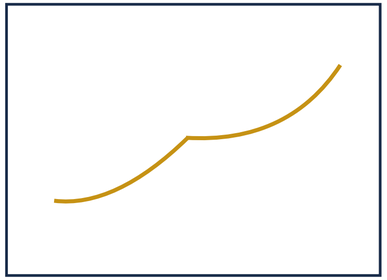

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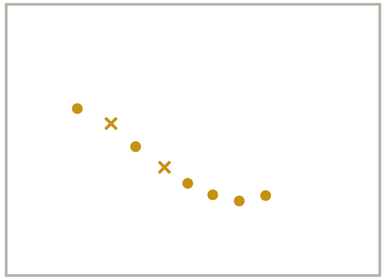
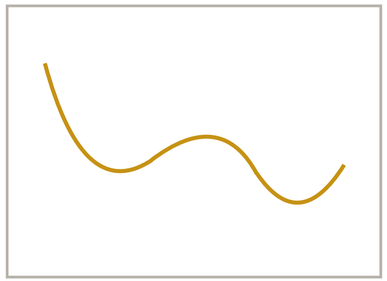


Model convergence
failures

Fixable using a
smooth transition
(e.g., a cubic function)

Gradient-based optimization has several challenges but each challenge is addressable



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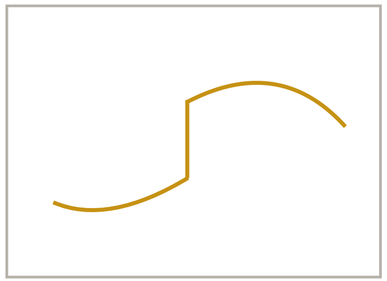

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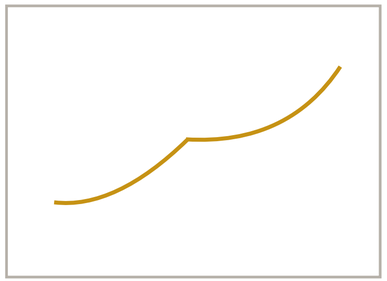

Non-differentiable
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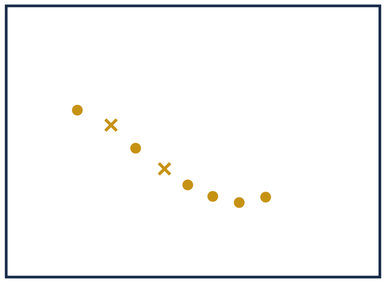

Model convergence
failures

Can handle failures
with a backtracking
line search

Improved

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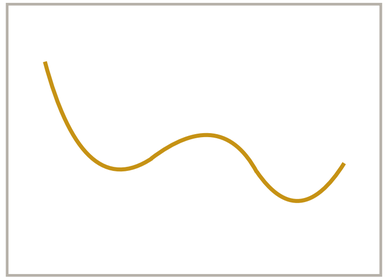

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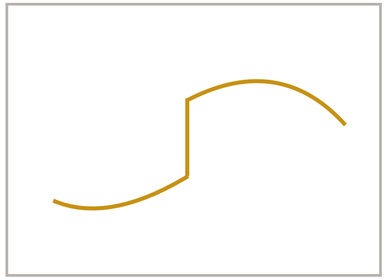

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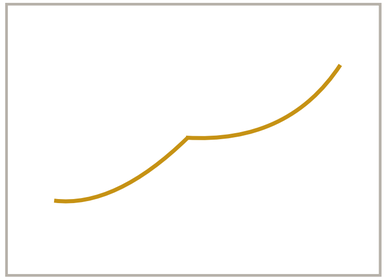

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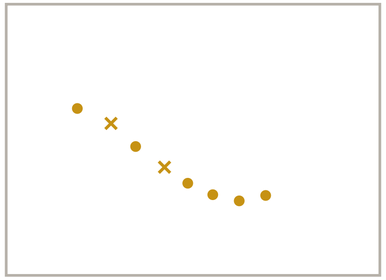

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Can handle failures with a backtracking line search

Gradient-based optimization

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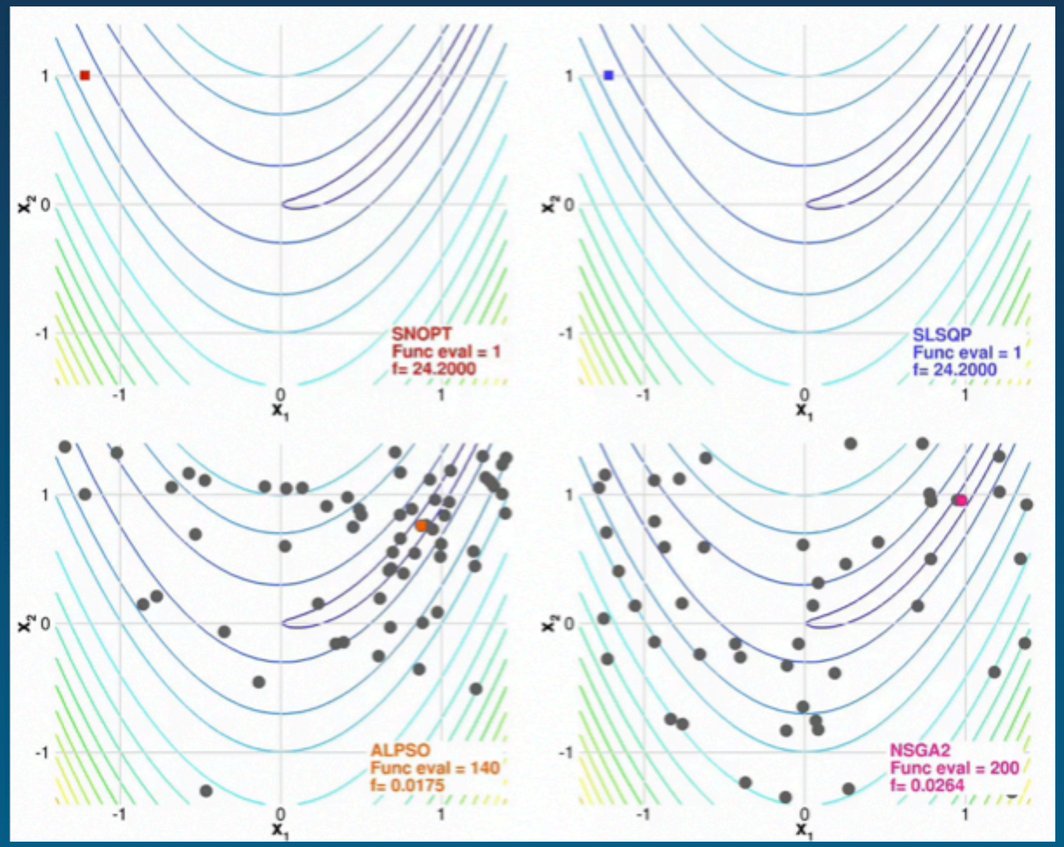

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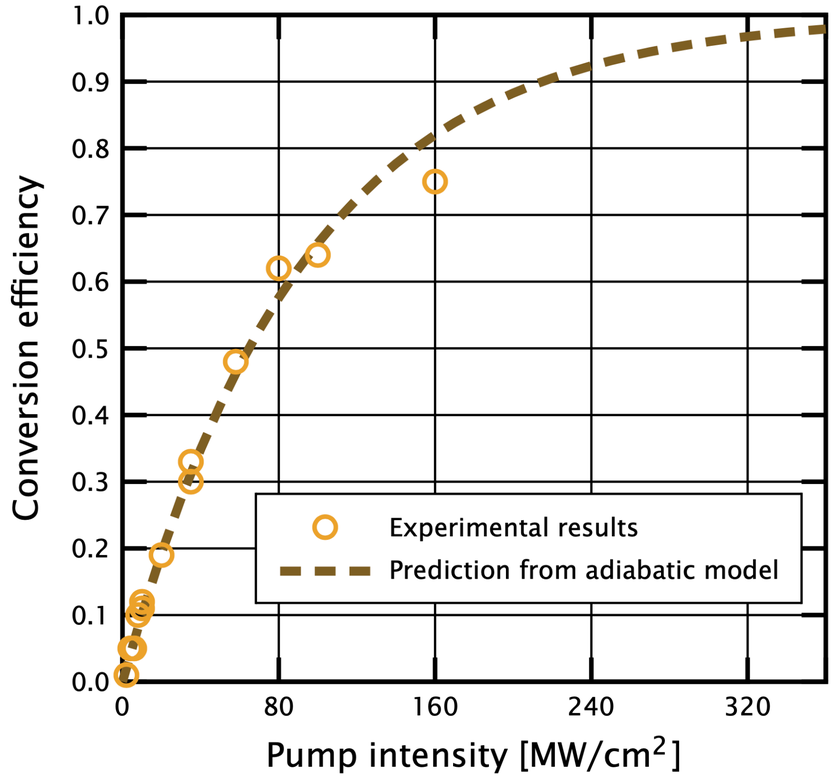
However, in practice, these challenges do not diminish the value of gradient-based optimization.

Original

Efficiency of adiabatic frequency conversion

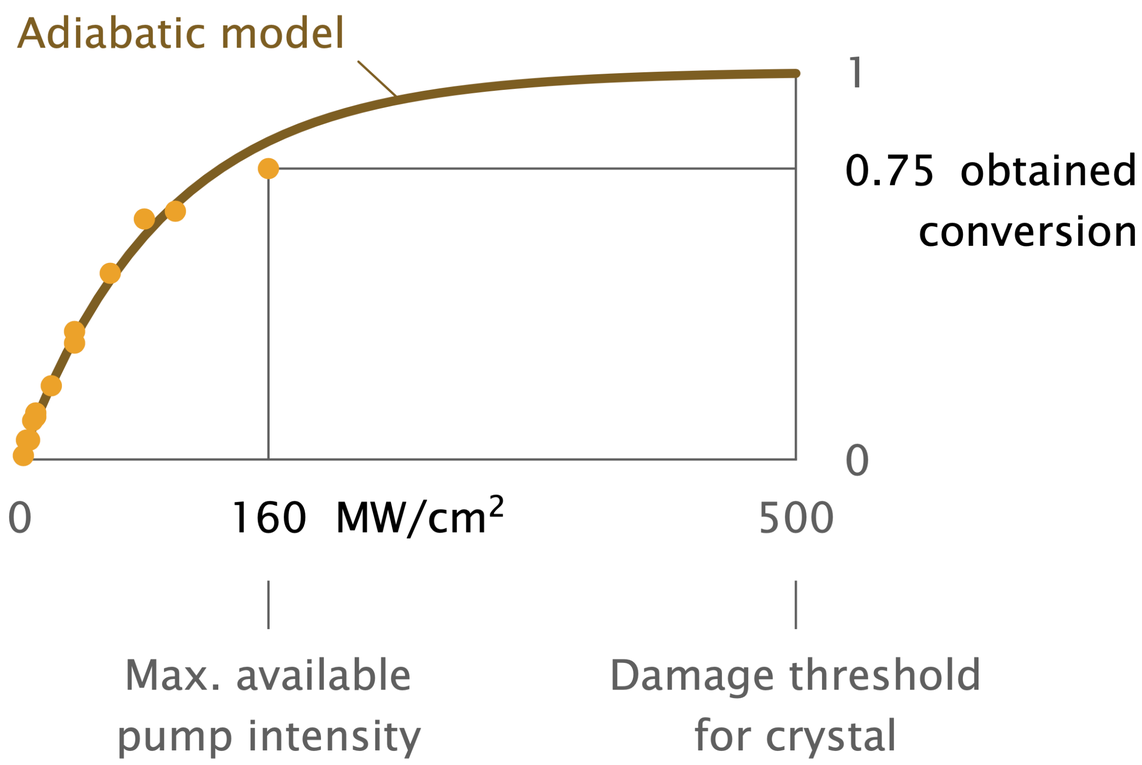


- Maximum pump intensity available experimentally: 160 MW/cm^2
- λ_1 = 1530 nm; λ_2 = 1064 nm (Q-switched Nd:YLF)
- The maximum demonstrated conversion efficiency was 75 percent
- Periodically poled crystal can get damaged from 500 MW/cm^2 of pump intensity

["Traditions, templates, and group leaders", Jean Luc Doumont]

Improved

The conversion approaches 100% for high enough pump intensity



[“Traditions, templates, and group leaders”, Jean Luc Doumont]

The ability to prepare and delivery effective technical presentations is an essential skill

Technical presentations are a regular part of most engineers’ work lives

Your ideas are only as good as your ability to communicate them

With the right approach, you can greatly improve your presentation skills

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Your ideas are only as good as your ability to communicate them

With the right approach, you can greatly improve your presentation skills

- Focus on the process
- Don't be afraid to try new things and make mistakes
- Make sure to put sufficient effort into the planning process
- With consistent effort into each presentation you give, *your skills will improve significantly over time*

Effective technical presentations

Motivation

Presentation design

- Plan and design your presentation
- Be consistent
- Make the presentation structure readily apparent

Slide design

- Maximize signal-to-noise ratio
- Have a single message each slide
- Make sure your slides are well-polished

Delivery

Effective technical presentations

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Delivery

Plan and design your presentation

Gather all the content you have for the presentation (figures, tables, lists, etc.)

Think about who you are presenting to, what you are presenting, and why

Have multiple brainstorming sessions, create outlines, and iterate

Your final outline should list all the slides: formulate one message per slide

Example outline

Attention grabber
Motivation
Main message

Section 1
⋮
Section 2
⋮
Section 3
⋮

Summary
Conclusion / takeaway

Consistency reduces cognitive load and improves professionalism

Font type—I recommend a sans serif font

Font sizes—stick with a small number of sizes across slides

Colors—stick with 3~4 colors across slides

Margins—stick with roughly the same-size border margins and space between figures, text, etc.

Slide layout—I recommend a 16:9 aspect ratio, two-column layout for most slides

Slide template—I recommend a clean, minimalist template with white background

Make the presentation structure readily apparent

Give structural cues to the audience, but not in an overpowering way

- Show a presentation outline once at the beginning
- For longer presentations, briefly show the outline between major sections
- Another effective approach is to show the subsection heading with a solid background of a color that sharply contrasts (e.g., switch from light to dark)
- Showing an outline “bar” on every slide is not a high-value use of space

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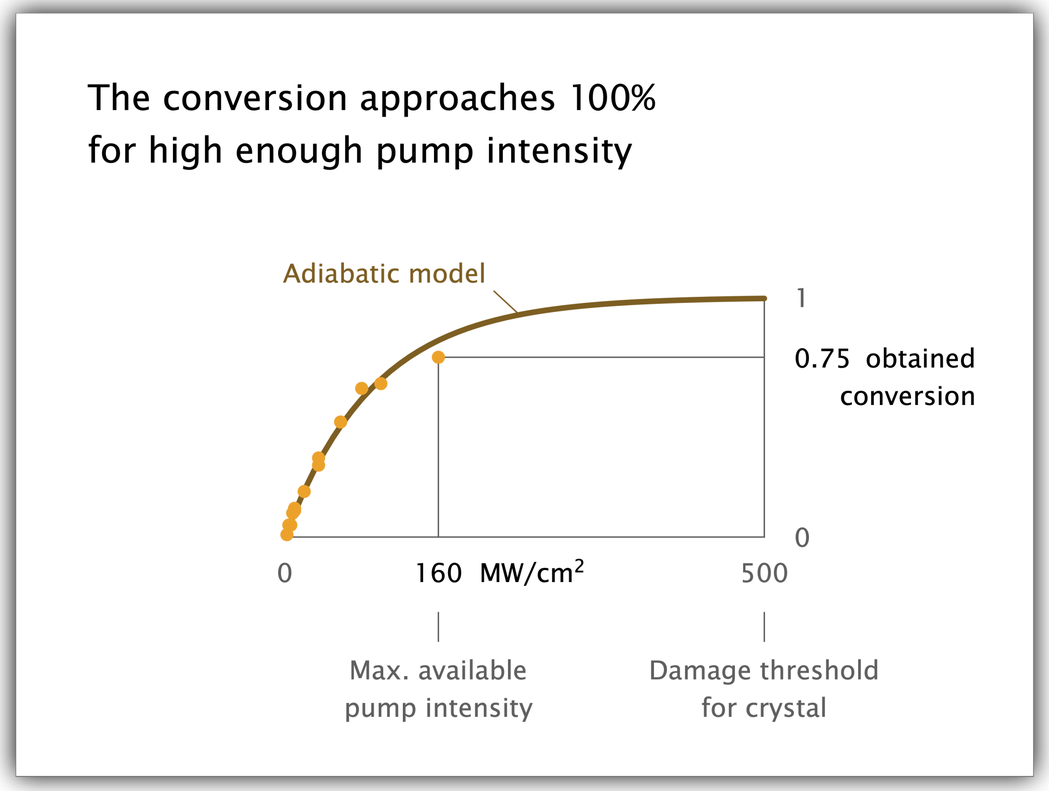
Delivery

Maximize signal-to-noise ratio

Signal = message you are conveying

Noise = everything else



[“Traditions, templates, and group leaders”, Jean Luc Doumont]

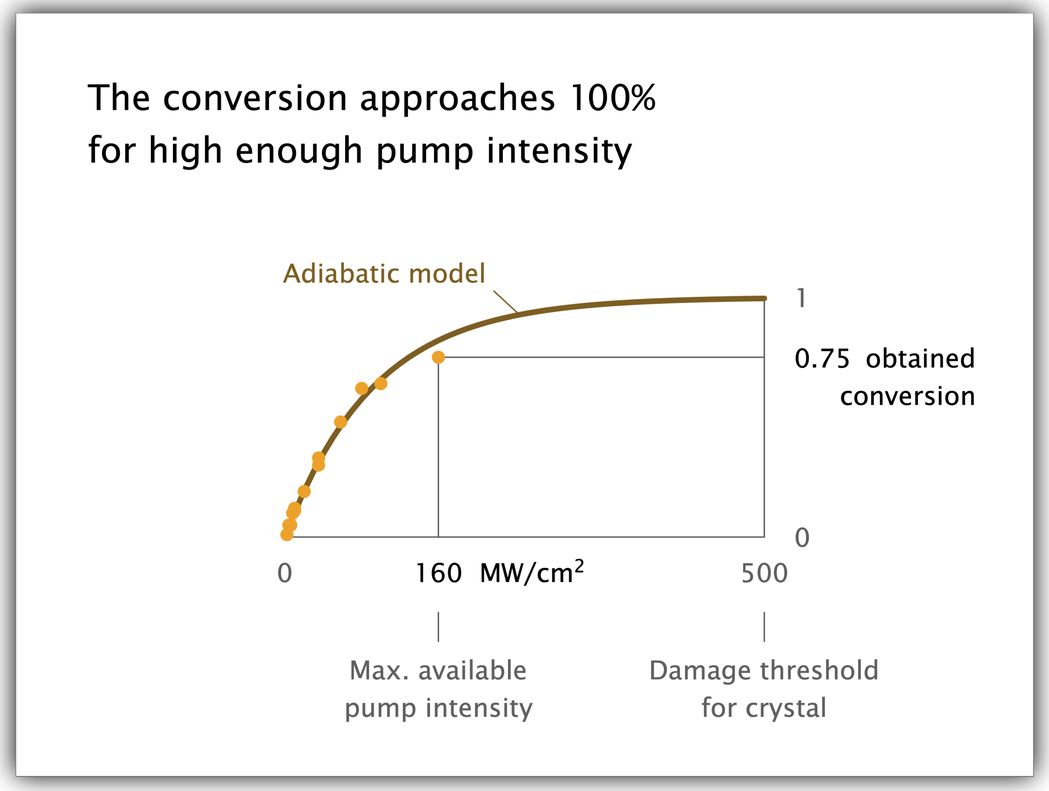
Plan a single message per each slide

Try to identify one main message you are trying to convey through the slide

Make this the title of your slide, with an appropriate line break

The contents of the slides should provide supporting evidence for this message

If you list the titles of your slides in order, they should tell a logical, compelling story



[“Traditions, templates, and group leaders”, Jean Luc Doumont]

Make sure your slides are well-polished

Try to align all text, figures, tables, and other content as much as possible in a grid

Balance all margins and spaces uniformly between text, figures, tables, and so on

Use vector graphics for equations and figures whenever possible—when not possible, use a very-high-resolution screenshot so that pixels are not visible

Many images have content in the foreground, in front of a white background; for this reason, it is preferable to use a white background for your slides as well

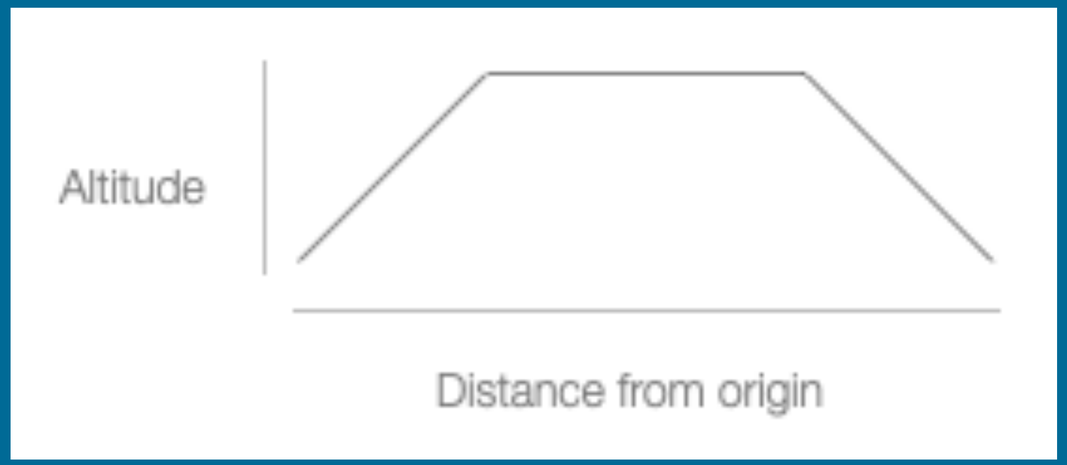
Annotate figures to highlight where you want the audience to focus their attention and what the takeaway message is

Original slide

The electric range equation relates range (a requirement), battery energy density (a technology parameter), lift-to-drag ratio (aerodynamic efficiency), battery weight fraction (structural efficiency), and propulsion system efficiency.

$$R = \frac{\eta}{g} \frac{E}{m_b} \frac{L}{D} \frac{W_b}{W_0}$$

Mission profile

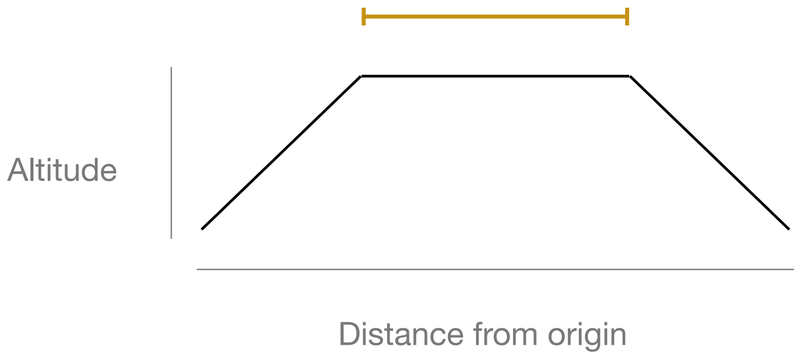


Improved slide

Electric range equation

$$R = \frac{\eta}{g} \frac{E}{m_b} \frac{L}{D} \frac{W_b}{W_0}$$

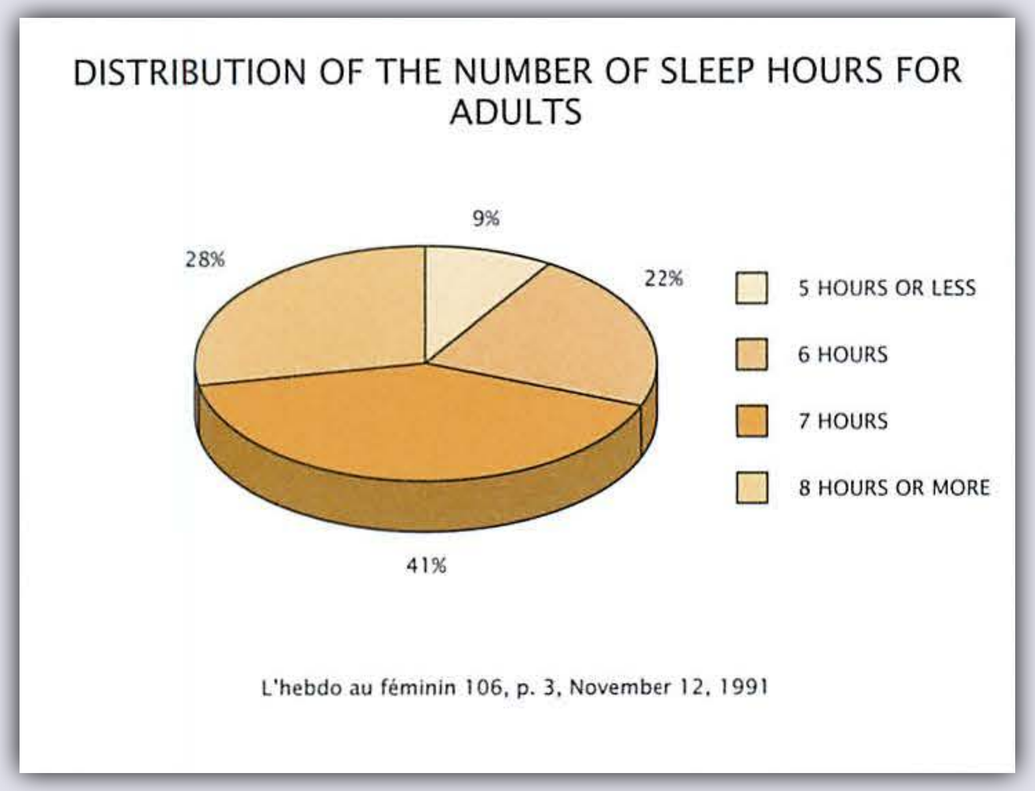
The range equation considers only cruise



It relates:

- Range
  - (requirement)
- Battery energy density
  - (technology parameter)
- Lift-to-drag ratio
  - (aerodynamic efficiency)
- Battery weight fraction
  - (structural efficiency)
- Propulsion system efficiency
  - (propulsive performance)

Original slide



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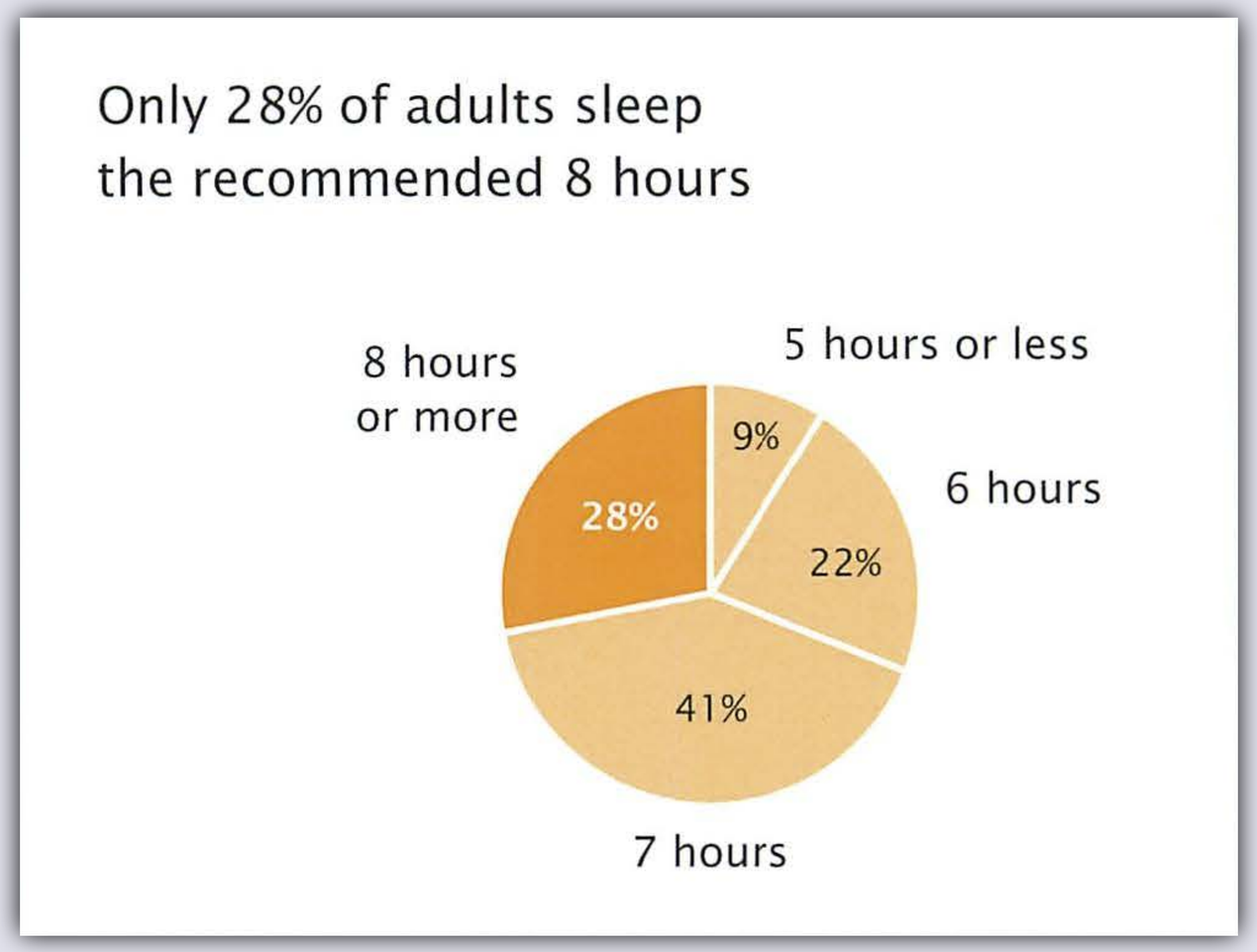
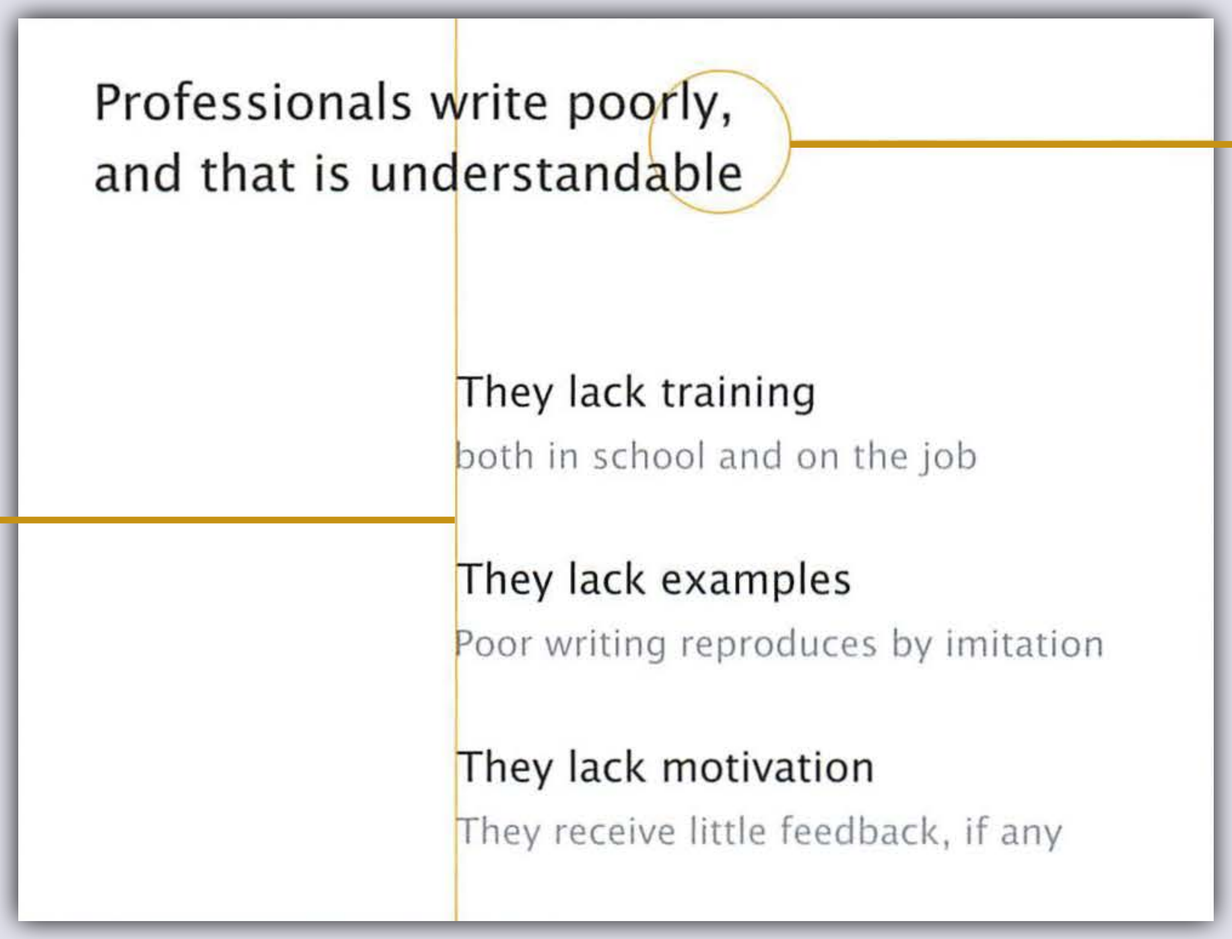


[“Trees, maps, and theorems”, Jean Luc Doumont]

Aim for structure and visual harmony

Put your message in the title, leaving space for two lines

Consider using an extra-wide left margin (rule of 3rds)

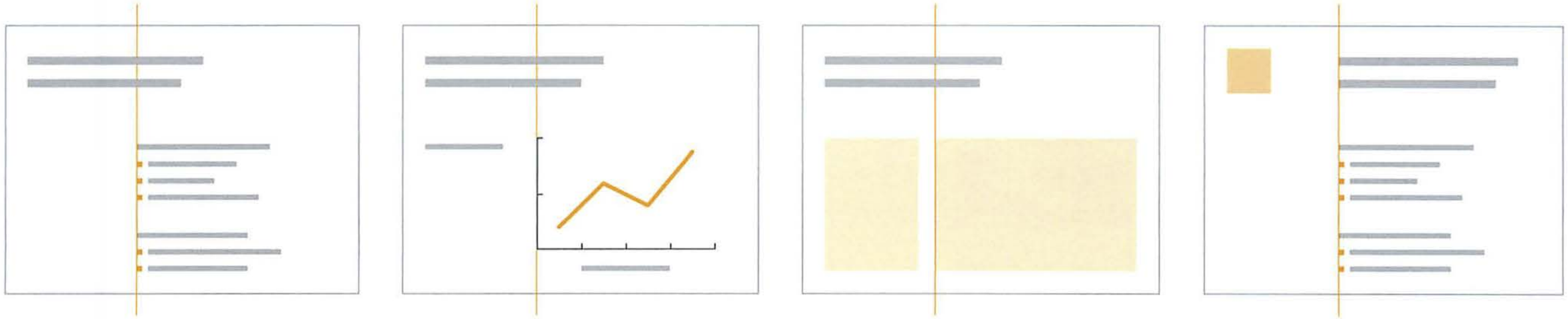


Select the line break strategically

Use spacing to clearly communicate the structure of the slides's messages

[“Trees, maps, and theorems”, Jean Luc Doumont]

Aim for structure and visual harmony



Displayed lists
aligned left, with square bullets

A single graph
aligned left, with label in margin

Two photographs
aligned far left, coordinated sizes

Displayed lists, with logo
all aligned left, including the title

[“Trees, maps, and theorems”, Jean Luc Doumont]

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Presentation design

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- Make the presentation structure readily apparent

Slide design

- Maximize signal-to-noise ratio
- Have a single message each slide
- Make sure your slides are well-polished

Delivery

Effective delivery is natural, audience-focused, and reinforces the slides

Practice 2~3 times, but no more because then you will start to memorize and your delivery may become unnatural

Do not read from a script—most often, this is obvious to the audience and has the effect of making your delivery less engaging and compelling

Being nervous is a good sign, because that means you care. To overcome it, try to focus on the audience and your goal of communicating to them, rather than yourself

The delivery and the slide should both be able to stand on its own—think of your delivery has reinforcing the message in the slide & vise versa (they are redundant)

If you have too much text in the slide, the audience will ignore you and read, or they will listen to you and the text is useless noise—both are undesirable

Action items

Try to put these ideas into practice for CoDR

Find and watch seminars on effective presentations (e.g., by Jean-Luc Doumont)

References:

https://www.principiae.be/book/barriers/

https://www.principiae.be/book/X0301.php

https://mitcommlab.mit.edu/meche/commkit/technical-presentation/